

The 8 Golden Rules of Interface Design

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https://youtu.be/INY_M3Ebtck?si=ZfVcA6OIsx5VIGLq

Shneiderman's Eight Golden Rules

Eight principles for creating user interfaces that are simple to understand and operate are known as Shneiderman's Eight Golden Rules. The rules are:

- **Strive for consistency:** Throughout the interface, use the same fonts, terms, and design components.
- **Cater to universal usability:** Create an interface that can be used by people with a variety of skill levels.
- **Offer informative feedback:** Give users feedback on what they're doing so they're aware of the situation.
- **Design dialogs to yield closure:** Make sure that users know when a dialog box is complete and what they need to do to proceed.
- **Prevent errors:** Design the interface to prevent errors from happening in the first place.
- **Permit easy reversal of actions:** Allow users to easily undo their mistakes.
- **Keep user in control:** Make users feel like they are in control of the system.
- **Reduce short-term memory load:** Reduce the amount of knowledge that users must retain in order to use the system.

1. Strive for consistency

Consistent sequences of actions should be required in similar situations;

- 在类似的情境中，用户执行的操作流程应该保持一致。
- 在电商网站中，无论是购买书籍还是衣服，流程都应是：选择 → 加入购物车 → 结账 → 确认。
- 符合用户习惯

Identical terminology should be used in prompts, menus, and help screens

- 如果按钮叫“Submit”，那帮助文档也应该说“Click Submit”，而不是“Click Send”。
- 在中英文界面中，“保存”应始终对应“Save”，不能一处叫“Store”。

Consistent color, layout, capitalization, fonts, and so on, should be employed throughout

- 颜色、布局、大小写、字体等视觉元素应保持一致。

Exceptions, such as required confirmation of the delete command or no echoing of passwords, should be comprehensible and limited in number

- 例外情况（如删除前确认、密码不显示）应易于理解且数量有限。
 - 删除文件时弹出“确认删除？”是合理的例外。
 - 密码输入框不显示字符是安全例外，但用户名输入框就不该隐藏。

2. Cater to universal usability

Recognize the needs of **diverse** users and design for plasticity, facilitating transformation of content.

Novice 初学者 to **expert** differences, **age** ranges, **disabilities**, **international variations**, and technological diversity each enrich the spectrum of requirements that guides design.

Adding features for **novices**, such as **explanations**, and features for **experts**, such as shortcuts and faster pacing 步调、节奏, enriches the interface design and improves perceived quality.

- **Novice to expert differences** (新手与专家差异)：新手需要引导与解释，专家则追求效率与控制。
 - 新手用户：提供课程导航、术语解释、学习进度提示、自动播放关闭 autoplay toggle 选项等
 - 专家：支持跳过基础模块、快速访问高级内容、提供快捷键控制视频播放速度 keyboard shortcuts for playback speed
- **Age ranges** (年龄层次)：年长者可能需要**更大字体**、简化流程 simplified workflow、色彩对比增强 enhanced color contrast；年轻人可能偏好探索性设计。
- **Disabilities** (残障情况)：包括视觉 (Screen reader support, color-blind-friendly palettes)、听觉 (Subtitles)、运动障碍 (voice control, enlarged clickable areas)，需提供辅助功能。
 - Enlarged clickable area – 扩大点击区域
 - 指的是将按钮、链接或交互元素的**可点击范围**设计得更大
 - 老年用户 (手指控制不精确)、运动障碍用户 (使用辅助设备或手势)
- **International variations** (国际差异)：语言、图标理解 (Culturally neutral icons)、颜色含义等文化因素。
 - 文化中性图标”是指**不依赖特定文化背景理解**的图标设计
 - 使用猪表示“储蓄”在某些文化中可能不适宜。
- **Technological diversity** (技术多样性)：设备性能、网络条件、操作习惯的差异。

Universal usability 强调的是：

“让所有用户都能有效、安全地使用系统，无论其年龄、能力、设备或环境。”

3. Offer informative feedback

For **every user action**, there should be an **interface feedback**.

For **frequent and minor actions**, the response can be modest 适中, whereas for **infrequent and major actions**, the response should be more substantial.

对于频繁且轻微的操作，可给予简洁反馈；而对于不常见且重要的操作，应提供更显著的反馈。

Visual presentation of the objects of **interest** provides a convenient environment for **showing changes explicitly**

- Visual presentation of the objects of interest 按钮、图表、进度条、卡片、列表项等界面元素。
- for showing changes explicitly 颜色变化、位置移动、数值更新、状态切换等。

关注于不断的反馈

- Button color change on hover
 - 输入错误: Show error message + red border
 - Icon color change or animation on click
 - 边框高亮、显示输入提示 Highlighted border, show input hint
 - 加载动画 – 表示系统正在处理: Loading spinner
 - 进度条: Progress bar
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4. Design dialogs to yield closure 设计对话流程以实现闭合感

Sequences of actions should be organized into groups with a **beginning, middle, and end**.

Informative **feedback at the completion of a group of actions** gives users the satisfaction of accomplishment, a sense of relief, a signal to drop contingency plans from their minds, and an indicator to prepare for the next group of actions.

For example, e-commerce websites => Select product → Add to cart → Checkout → Pay → Confirmation page

操作流程应有**明确起止**，结束时给予完成感反馈

让用户知道任务已完成，可安心进入下一阶段

如果可以明确看到，是阶段性任务的结束 => *Design dialogs to yield closure*

关注于一个阶段任务结束的反馈

- 完成注册流程: Welcome page with username
 - 成功登录: Show welcome message + redirect to dashboard
 - 表单填写完成并提交 'Submission successful' message + redirect to confirmation
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5. Prevent errors

As much as possible, design the interface so that **users cannot make serious errors**;

设计目标是“防错优先” (**Error Prevention**)，而不是“事后纠错”

不允许非法输入、不显示无效选项、不允许误操作。

- For example, **gray out menu items that are not appropriate and do not allow alphabetic characters in numeric entry fields**.
- Gray out, disable click 置灰处理，禁止点击

- Show 'Password must include uppercase, lowercase, and digits'

If users make an error, the interface should offer **simple, constructive, and specific instructions for recovery**.

- For example, users should **not have to retype an entire** name–address form if they enter an invalid zip code but rather should be guided to **repair only the faulty part**. 局部纠错
- Preserve filled data, guide to fix specific fields

Erroneous actions should leave the interface state unchanged, or the interface should give instructions about restoring the state.

错误操作不应改变界面状态，或界面应提供恢复状态的指导

- 如果用户误点'删除'，应弹出确认框，而不是立即清除数据。
- 如果状态已变，应提供“撤销”或“恢复”选项。

6. Permit easy reversal of actions

As much as possible, actions should be **reversible**.

This feature relieves anxiety, since users know that errors can be undone, and encourages exploration of unfamiliar options.

The units of reversibility may be a single action, a data–entry task, or a complete group of actions, such as entry of a name–address block.

- Support **Undo/Redo** 撤销/重做 buttons

7. Keep users in control

Experienced users strongly desire the sense that they are **in charge of the interface** and that the interface responds to their actions. -> 页面听话

- Customizable layout
- Skipping animations -> Provide 'Skip' button to avoid forced wait
- Support **keyword** search, filters, sorting => 用户主动获取信息，系统响应其意图
- Users can click arrows or swipe to change slides => 用户决定何时切换内容，而不是被动等待自动播放

They don't want surprises or changes in familiar behavior,

- 当界面设计与用户以往的生活习惯不一致时，如果处理不当，会导致用户失去控制感、产生挫败感，甚至放弃使用
 - 用户习惯从右向左阅读（如阿拉伯语用户），但界面为左到右布局

- 用户习惯点击图标展开菜单，但系统要求长按

and they are annoyed by **tedious data-entry sequences**, difficulty in obtaining necessary information, and **inability to produce their desired result**.

- Users cannot clearly read text, losing control over information
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8. Reduce short-term memory load

Humans' limited capacity for information processing in short-term memory (the rule of thumb is that people can remember 'seven plus or minus two chunks' of information 经验法则是人们能记住“七个上下两个信息块”)

It requires that designers **avoid** interfaces in which **users must remember information from one display and then use that information on another display**.

It means that cellphones should not require reentry of phone numbers,

这意味着手机不应要求用户重新输入电话号码 – 复制粘贴、自动填充、联系人选择等方式可以避免重复输入。

Website locations should remain visible

用户不应被迫记住网址，应能随时查看或点击链接。

And lengthy forms should be compacted to fit a single display. 避免分页表单或跳转式填写，减少用户在不同页面之间记忆字段内容。